Year 12 Further Maths - Matrices

Similar questions - Determinants, area and volume scale factors

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A transformation T is represented by $\mathbf{T} = \begin{pmatrix} x^2+2 & -3 \\ 2-x^2 & x^2+4 \end{pmatrix}$. A quadrilateral Q of area 8 units is transformed by T to Q'. Find the smallest possible area of Q'. **[5]**

2 A transformation is represented by $\mathbf{A} = \begin{pmatrix} x & 1 & 0 \\ 2 & x-1 & 3 \\ 0 & 1 & x \end{pmatrix}$. A tetrahedron H has vertices O, P, Q and R and volume 6 units; P', Q', R', H' are the images under $\mathbf{A}$.

(a) For $x = 3$, find the volume of H' and determine whether $\mathbf{A}$ preserves the orientation of H. **[3]**

(b) Find the values of x for which O, P', Q', R' are coplanar. **[4]**

3

(a) The matrix $\mathbf{A} = \begin{pmatrix} t & 6 \\ 1 & t-5 \end{pmatrix}$. Find, in fully factorised form, $\det \mathbf{A}$ in terms of t, and state the values of t for which $\mathbf{A}$ is singular. **[4]**

(b) A system of equations is $x+y+bz=1$, $x+by+z=1$, $bx+y+z=1$, where b is a real number. Determine, with justification, the values of b for which the system has (i) a unique solution, (ii) infinitely many solutions, (iii) no solution. **[7]**

4 In this question you must show detailed reasoning. The matrix $\mathbf{A} = \begin{pmatrix} a & -4 & a-2 \\ a+6 & a & 3 \\ 0 & -9 & a-1 \end{pmatrix}$ where a is a constant. Find all possible values of a for which $\det \mathbf{A}$ has the same value as when $a = 2$. **[6]**

Question 1

$\det \mathbf{T} = (x^2+2)(x^2+4)+3(2-x^2) = x^4+3x^2+14 \geqslant 14$, with equality when $x=0$. Area scale factor $= |\det \mathbf{T}| \geqslant 14$, so smallest area of $Q' = 14 \times 8 = 112$ units

Question 2

(a) $\det \mathbf{A} = x^3 - x^2 - 5x = 27 - 9 - 15 = 3 > 0$, so $V(H') = 3 \times 6 = 18$ units and orientation is preserved

(b) Coplanar when $\det \mathbf{A} = 0$: $x(x^2 - x - 5) = 0$, $x = 0$ or $x = \dfrac{1 \pm \sqrt{21}}{2}$

Question 3

(a) $\det \mathbf{A} = t(t-5) - 6 = t^2 - 5t - 6 = (t-6)(t+1)$; singular for $t = 6$ and $t = -1$

(b) The determinant of the coefficient matrix is $-(b-1)^2(b+2)$. (i) Unique solution when $b \neq 1$ and $b \neq -2$. (ii) $b = 1$: all three equations are $x+y+z = 1$, so infinitely many solutions. (iii) $b = -2$: adding the equations gives $0 = 3$, so no solution

Question 4

Expanding gives $\det \mathbf{A} = a^3 - 6a^2 + 11a + 84$; at $a = 2$ this is 90. Solve $a^3 - 6a^2 + 11a - 6 = 0$: $(a-1)(a-2)(a-3) = 0$, so $a = 1, 2, 3$